SPECIFICATIONS — PROJECT	
Project title	**Collaborative Fog Computing for Real-Time Predictive Maintenance, Dynamic Task Scheduling and Fault-Tolerant Industrial Monitoring.** Team: 3 students — Cloud / Fog / IoT / AI / DevOps project.
Targeted profession / domain	Industrial maintenance in a production plant. The goal is to continuously monitor machines and detect early warning signs of failure. Simulated equipment: electric motors, pumps, compressors, industrial fans, conveyors, CNC machines (main focus: electric motors and rotating machines, whose vibration, temperature, speed and current allow realistic fault simulation).
Context	A modern plant produces continuous data (temperature, vibration, pressure, speed, current, power, frequency, state). A classic Cloud-only architecture causes: • large data volume and high bandwidth consumption; • latency and slow reaction to an anomaly; • dependency on the Cloud and overload of the central server. Proposed solution: a collaborative Cloud–Fog architecture where data is first processed on Fog nodes close to the machines.
Problem statement	*How to design a collaborative Cloud–Fog architecture able to monitor industrial equipment in real time, detect failure precursors locally, dynamically distribute analysis tasks among several Fog nodes, and keep the service running when a node is overloaded or fails?* Four problems to solve simultaneously: • detect anomalies quickly; • reduce dependency on the Cloud; • distribute tasks intelligently between Fog nodes; • maintain the service when a Fog node becomes unavailable.

General objective	Develop a virtual predictive-maintenance platform able to: • simulate 12 industrial machines (3 zones A/B/C, one Fog node per zone) producing realistic data every second; • transmit data via MQTT and analyse it locally on Fog nodes (cleaning, feature extraction, anomaly detection, failure prediction); • balance load: migrate non-critical tasks from an overloaded Fog to the others, critical tasks stay local; • tolerate Fog failure (heartbeat, task recovery, node re-integration); • send key information to the Cloud (history, ML training, global analytics) and display the factory state on a dashboard.